

Software Requirements Specification (SRS)

Rotana Spa Management System

Document	Software Requirements Specification
Project	Rotana Spa Management System
Version	1.0
Status	Approved
Prepared for	Business owner / client
Target market	Ethiopian spa & gym operators (Addis Ababa)

1. Introduction

1.1 Project Name

Rotana Spa Management System

1.2 Purpose

The Rotana Spa Management System is a web application that digitises the day-to-day operations of a spa and gym: member registration and membership plans, visits and treatments, therapist and trainer workloads, inventory and towels, facilities and gates, service orders handed to the cashier, and operational reports.

1.3 Scope

The system covers the following functional areas:

- **Customers** — member profiles, medical records, loyalty, visit history.
- **Membership** — plans, member registration, renewals, freeze/transfer, digital (RFID) cards, QR access passes.
- **Operations** — reception visits, appointments, sessions, queue, customer requests, towel management, and **service orders**.
- **Spa** — treatment catalogue (services), therapists, treatment rooms, bookings, packages.
- **Gym** — trainers, workout plans, fitness assessments, body measurements, classes, gym attendance.
- **Inventory** — products, consumables, stock usage, suppliers.
- **Staff** — employees, schedules, commission, performance.
- **Facilities** — rooms, lockers, equipment, maintenance, entry gates.
- **Reports** — membership, attendance, service orders, therapist, trainer, inventory.
- **Administration** — users, roles and permissions, system settings, audit logs, companies, demo licenses, notifications.

Architecture boundary — Sales/POS: This application is an **operational** system. It does **not** create invoices, calculate prices, discounts or tax, accept payment, or connect to the separate Sales/POS database. Finishing a treatment produces an **80 mm draft Service Order** (visit, customer, therapist, service lines and quantities only) that the customer hands to the cashier, who completes the financial transaction in the separate POS application.

1.4 Objectives

- Provide one place for front-desk, therapist, trainer and management workflows.
- Replace manual/paper service tracking with a reliable, searchable record.
- Enforce role-based access control and keep a full audit trail.
- Produce operational reports without duplicating POS financial data.
- Support English and Amharic interfaces and Ethiopian Birr (ETB) currency.

1.5 Definitions and Abbreviations

Term	Meaning
Visit	A reception check-in that carries a customer through check-in → assigned → in-treatment → finished.
Service order	A non-financial, draft list of treatment services and quantities generated when a visit finishes.
Member	A registered customer with a membership plan, card and/or QR pass.
RFID card	Digital membership card used at gates / gym.
QR pass	QR-based access pass for member or guest entry.
Rate card	The configured price of a service, session, day pass, facility or membership.
ETB	Ethiopian Birr, the default currency.
SPA-MGMT	The application's internal name for the licensed operations module.

2. System Overview

2.1 Spa operation workflow

Front desk	Therapist / Treatment	Cashier (POS, separate)
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Member/guest check-in	-> Assign therapist	-> Receives draft order
Visit created	Start treatment	Completes the payment
(SPA-000001)	Record services used	in the separate POS app
	Finish treatment	
	Generate service order (draft)	

Print 80mm thermal slip
Hand order to cashier

1. Reception checks a customer (member or guest) into a **visit**.
2. A therapist is assigned; the visit moves to **in treatment**.
3. The therapist records the services performed (with quantities).
4. On **finish**, the system generates a **draft service order** with a snapshot of the services.
5. The slip is **printed** (58/80 mm thermal) and **handed to the cashier**.
6. The visit status becomes **handed_to_cashier**; the financial transaction happens in POS.

2.2 User roles

Role	Description
super_admin	Platform owner. Manages companies, demo licenses, audit logs, manuals. Cannot create tenant records directly (no company).
admin	Company administrator. Full access to the company's data, users, roles, settings.
manager	Company manager. Operational access and approvals; limited administrative actions.
receptionist	Front desk: check-ins, appointments, registrations, queue, towels, service orders.
guest	Limited, view-only access.
therapist	Created via the Roles screen where needed; assigned treatment workflows.

Permissions are enforced per **resource** and **action** (`view` , `create` , `edit` , `delete` , `approve`) through `role_permissions` .

2.3 Business rules

- A visit number is generated per company from `spa_visit_counters` (`SPA-000001` , `SPA-000002` , ...).
- A service order number is generated as `SO-SPA-000001` , ... per company.
- Only one visit may exist per appointment.
- Services may be added to a visit only after a therapist is assigned and treatment has **started**.
- A closed visit (`finished` , `order_printed` , `handed_to_cashier` , `cancelled`) cannot accept treatment changes.
- A service order can only be handed to the cashier after the visit is finished.
- Member `end_date` is required and defaults to 30 days when created without a plan.
- Super administrator must target an explicit `company_id` to create tenant records.
- Default currency is **ETB**; settings store company name, address, phone, email and currency.

3. Functional Requirements

Requirements are numbered `FR-n` with a priority of High (H), Medium (M) or Low (L).

3.1 Authentication & Users

ID	Requirement	Priority
FR-001	The system shall authenticate users by email and password.	H
FR-002	Passwords must be stored hashed (never plain text).	H
FR-003	The system shall issue a JWT on login and accept it as <code>Authorization: Bearer <token></code> .	H
FR-004	The system shall offer role-based sign-in entry points: company admin, super admin, employee, guest.	M
FR-005	An authenticated user's visible menu and pages must reflect their role permissions.	H

3.2 Customer management

ID	Requirement	Priority
FR-010	The system shall register a customer as a member with name, phone, email, address, plan and start date.	H
FR-011	The system shall generate a unique customer code (<code>MEM-001</code> , ...).	H
FR-012	The system shall search customers by name, code, phone, and view visit history.	H
FR-013	The system shall record medical records, loyalty tier/points and follow-ups.	M
FR-014	The system shall display a member's plan, status (active/expired) and balance dates.	H

3.3 Membership

ID	Requirement	Priority
FR-020	The system shall manage membership plans (spa, gym, cafe, general) with duration and active flag.	H
FR-021	The system shall create subscriptions linking member + plan with start/end dates and billing cycle.	H
FR-022	The system shall support renewals and freeze/transfer requests.	M

ID	Requirement	Priority
FR-023	The system shall manage RFID digital cards (active/inactive/lost/expired).	M
FR-024	The system shall issue QR passes with token, expiry, and max-uses tracking.	M
FR-025	The system shall manage day tickets and rate cards (service, session, day pass, facility, membership).	M

3.4 Operations — visits, appointments, service orders

ID	Requirement	Priority
FR-030	The system shall check a customer into a visit and generate a visit number.	H
FR-031	The system shall assign an active therapist to a visit.	H
FR-032	The system shall start and finish treatment, tracking timestamps.	H
FR-033	The system shall record services used per visit with quantity.	H
FR-034	On finish, the system shall generate a draft service order with a snapshot of service lines.	H
FR-035	The system shall support re-printing and handing the order to the cashier.	H
FR-036	The system shall create appointments with room/facility, service, start/end.	M
FR-037	The system shall manage a queue and customer requests.	L
FR-038	The system shall track towel issues and returns per visit.	M

3.5 Spa catalogue

ID	Requirement	Priority
FR-040	The system shall maintain a service catalogue (name, category, duration).	H
FR-041	The system shall maintain therapist profiles (specialties, certifications, commission).	H
FR-042	The system shall manage treatment rooms and spa facilities.	M
FR-043	The system shall manage packages.	L

3.6 Gym

ID	Requirement	Priority
FR-050	The system shall maintain trainer profiles.	H
FR-051	The system shall build workout plans per member (goal, sessions/week, exercises).	M

ID	Requirement	Priority
FR-052	The system shall record fitness assessments and body measurements (BMI auto-calculated).	M
FR-053	The system shall schedule classes with trainer, capacity and enrolment; derive full/open status.	M
FR-054	The system shall record gym check-ins.	M

3.7 Inventory

ID	Requirement	Priority
FR-060	The system shall maintain products and consumables with quantity and reorder level.	M
FR-061	The system shall auto-derive stock status (in-stock / low-stock / out-of-stock).	M
FR-062	The system shall record stock usage against services/visits.	L
FR-063	The system shall maintain suppliers.	L

3.8 Towels

ID	Requirement	Priority
FR-070	The system shall issue towels per visit (quantity, type).	M
FR-071	The system shall record returns and derive status (issued / partially-returned / returned / lost).	M

3.9 Printing

ID	Requirement	Priority
FR-080	The system shall print the service order as an 80 mm (or 58 mm) thermal slip.	H
FR-081	The slip shall show visit, customer, therapist, service lines and quantities — no prices.	H
FR-082	The system shall track <code>print_count</code> and <code>printed_at</code> per order.	M

3.10 Reports

ID	Requirement	Priority
FR-090	The system shall provide a Membership report (status, plans, subscriptions).	H
FR-091	The system shall provide an Attendance report (check-in volume, active visits).	H
FR-092	The system shall provide a Service Orders report (draft/printed/handoff volume, item counts).	H
FR-093	The system shall provide Therapist and Trainer reports (volume, usage).	M
FR-094	The system shall provide an Inventory report (stock position, low stock, usage).	M
FR-095	Reports shall filter by date range and company.	H

3.11 Administration & notifications

ID	Requirement	Priority
FR-100	The system shall manage users and assign roles.	H
FR-101	The system shall manage roles and per-resource permissions (view/create/edit/delete/approve).	H
FR-102	The system shall maintain company settings (name, address, phone, email, currency).	H
FR-103	The system shall log every create/update/delete to an audit trail with user, IP and old/new values.	H
FR-104	The system shall support notifications per user (read/unread).	M
FR-105	The super admin shall manage companies, demo licenses and audit logs.	H

3.12 Internationalisation

ID	Requirement	Priority
FR-110	The system shall support English and Amharic.	M
FR-111	The system shall allow switching the UI language.	M

4. Non-functional requirements

4.1 Usability

- NFR-001: All primary workflows (check-in, add service, finish, print) must be reachable within two clicks from the dashboard.
- NFR-002: UI language and currency must be configurable.

4.2 Performance

- NFR-010: Common list/read pages must respond within ~1s on a typical connection.
- NFR-011: The system must support concurrent front-desk and therapist usage per company.

4.3 Security

- NFR-020: All API routes except login require a valid JWT.
- NFR-021: All queries must be parameterised (SQL-injection safe).
- NFR-022: Tenant data access must be scoped by `company_id`; users may only see their own company's data.
- NFR-023: Passwords must be hashed with a strong algorithm (bcrypt).

4.4 Reliability

- NFR-030: Audit logging must not block the primary transaction.

- NFR-031: Migration scripts must be idempotent where possible (`IF NOT EXISTS`).

4.5 Compatibility

- NFR-040: Must run on modern Chrome, Edge and Firefox.
 - NFR-041: Must support 80 mm and 58 mm thermal receipt printers.
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5. Constraints and assumptions

- **Assumption:** A separate Sales/POS application handles all pricing, tax and payment.
 - **Constraint:** Super administrator cannot insert tenant rows without supplying a company.
 - **Constraint:** Deployed on PostgreSQL 15+ (managed Neon database in this environment).
 - **Assumption:** Data is stored per company; multi-tenant isolation is enforced in queries.
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6. Acceptance criteria (summary)

1. A receptionist can check in a customer, assign a therapist, start treatment, record services and finish.
 2. Finishing generates a draft service order that can be printed and handed to the cashier.
 3. A member can be registered, assigned a plan and issued an RFID/QR pass.
 4. All 5 roles see only the menu items and actions they are permitted to use.
 5. Every create/update/delete appears in the audit log.
 6. All six reports render for the selected date range with correct counts.
 7. The interface renders correctly in English and Amharic.
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7. Reference screens

See the companion documents:

- `PAGE_GUIDE.md` — screen-by-screen description.
- `USER_MANUAL.md` — step-by-step operator guide.
- `API_DOCUMENTATION.md` — API reference.
- `DATABASE_DESIGN.md` — database schema.